

A Family of Difference Sets in Non-cyclic Groups

ROBERT L. MCFARLAND

Route 8, Lexington, Ohio

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A construction is given for difference sets in certain non-cyclic groups with the parameters $v = q^{s+1}[(q^{s+1} - 1)/(q - 1)] + 1$, $k = q^s (q^{s+1} - 1)/(q - 1)$, $\lambda = q^s (q^s - 1)/(q - 1)$, $n = q^{2s}$ for every prime power q and every positive integer s . If q^s is odd, the construction yields at least $\frac{1}{2}(q^s + 1)$ inequivalent difference sets in the same group. For $q = 5$, $s = 2$ a difference set is obtained with the parameters $(v, k, \lambda, n) = (4000, 775, 150, 625)$, which has minus one as a multiplier.

1. INTRODUCTION

Let G be a finite group of order v . A subset D of order k is a *difference set* in G with parameters (v, k, λ, n) in case every non-identity element g in G can be expressed in exactly λ ways as

$$g = d^{-1} d',$$

with d and d' in D . The parameter n is defined by

$$n = k - \lambda.$$

A good discussion of difference sets can be found in each of the books Hall [4], Mann [5], and Ryser [13].

Let q be a prime power and let s be a positive integer. We construct difference sets with the parameters

$$\begin{aligned} v &= q^{s+1} \left(\frac{q^{s+1} - 1}{q - 1} + 1 \right), \\ k &= q^s \left(\frac{q^{s+1} - 1}{q - 1} \right), \\ \lambda &= q^s \left(\frac{q^s - 1}{q - 1} \right). \\ n &= q^{2s}, \end{aligned} \tag{1.1}$$

in a group G which is the direct product

$$G = E \times K,$$

where E is an elementary Abelian group of order q^{s+1} and K is any group of order $[(q^{s+1} - 1)/(q - 1)] + 1$. If q^s is odd, the construction yields at least $\frac{1}{2}(q^s + 1)$ *inequivalent* difference sets in the same group. Gordon, Mills, and Welch [2] were the first to show that, for every positive integer i , there exists a set of i inequivalent difference sets with the same parameters in some group; however, their construction was for difference sets in cyclic groups. The construction also yields a difference set with the parameters

$$(v, k, \lambda, n) = (4000, 775, 150, 625)$$

which has minus one as a *multiplier*. This is the first example of a difference set having minus one as a multiplier whose parameters are *not* of the form

$$(1.2) \quad (v, k, \lambda, n) = (4m^2, 2m^2 - m, m^2 - m, m^2)$$

for some integer m . Note that we obtain difference sets with parameters of the form (1.2) for $q = 2$ and every s .

2. THE CONSTRUCTION

Let q be a prime power and let s be a positive integer. Let $F = \text{GF}(q)$ be a finite field of order q and let V be a vector space of dimension $s + 1$ over F . Let $H_1, \dots, H_r$ be the hyperplanes of V , i.e., subspaces of dimension s . The number r of hyperplanes is also the number of one-dimensional subspaces since the orthogonal subspace of a hyperplane is a one-dimensional subspace, and conversely. There are $q^{s+1} - 1$ non-zero elements in V and each of them generates a one-dimensional subspace over F containing $q - 1$ non-zero elements. Thus

$$(2.1) \quad r = \frac{q^{s+1} - 1}{q - 1}.$$

Let E be the additive group of V and consider $H_1, \dots, H_r$ as subgroups of E . Let $e_1, \dots, e_r$ be any r elements of E , not necessarily distinct. Let K be any group of order $r + 1$, not necessarily Abelian, and let $k_1, \dots, k_r$ be r distinct elements of K . Let $(H_i + e_i, k_i)$ denote the coset of H_i in the group $G = E \times K$ that contains the element (e_i, k_i) . We prove that

$$(2.2) \quad D = \{(H_i + e_i, k_i) \mid i = 1, \dots, r\}$$

is a difference set in G with the parameters (1.1). Note that the order of G is $v = q^{s+1}(r+1)$ and the number of elements in D is $k = q^s r$.

3. THE VERIFICATION

The collection of subgroups $H_1, \dots, H_r$ clearly contains every element of E . The identity element is replicated r times; we assert that every non-identity element is replicated exactly $(q^s - 1)/(q - 1)$ times. For, given any two non-identity elements of E , there is an automorphism that interchanges them. But every automorphism of E permutes the H_i 's among themselves. Thus every non-identity element is replicated the same number of times. A counting argument yields $(q^s - 1)/(q - 1)$ as the common number of replications.

Clearly, each element of H_i is replicated exactly q^s times in the sum

$$H_i + H_i = \{a + b \mid a, b \in H_i\}.$$

We assert that if $H_i \neq H_j$, then each element of E is replicated exactly q^{s-1} times in the sum

$$H_i + H_j = \{a + b \mid a \in H_i, b \in H_j\}.$$

By a vector space dimensional argument, $H_i + H_j$ contains every element of E . If $e = a + b$ with a in H_i and b in H_j is one representation of an e in E , then every representation of e in $H_i + H_j$ is of the form

$$e = (a + h) + (b - h)$$

for some h in $H_i \cap H_j$. As a vector space, $H_i \cap H_j$ has dimension $s - 1$ over F . Therefore, $H_i \cap H_j$ has cardinality q^{s-1} . The assertion follows.

From now on we assume that the operation of the groups E, K, G is multiplication and we reserve “+” for the addition in the respective group rings ZE, ZK, ZG of these groups over the ring Z of rational integers. The identity elements of E, K, G will be denoted by $1_E, 1_K, 1_G$, respectively. We identify a subset S of the group E with the element

$$S = \sum_{e \in S} e$$

in ZE that has coefficient one on each element of S and coefficient zero on the other elements of E . In particular, E itself denotes the element in ZE with every coefficient one, $E - 1_E$ denotes the element that has every coefficient one except for a zero coefficient on 1_E , and a group element e in E denotes the element that has coefficient one on e and all other coefficients zero. Analogous conventions hold for the group rings ZK and ZG .

The above derived properties of the H_i 's can be expressed in the group ring notation for ZE as

$$(3.1) \quad H_1 + \cdots + H_r = r1_E + [(q^s - 1)/(q - 1)][E - 1_E] \\ = q^s 1_E + [(q^s - 1)/(q - 1)] E,$$

$$(3.2) \quad H_i^2 = q^s H_i,$$

$$(3.3) \quad H_i H_j = q^{s-1} E, \quad i \neq j.$$

Let $k, k_1, \dots, k_r$ be the $r + 1$ elements of K . Then, in the group ring notation for ZK ,

$$(3.4) \quad \sum_{i \neq j} k_i^{-1} k_j = (K - k^{-1})(K - k) - \sum_{i=1}^r k_i^{-1} k_i \\ = K^2 - 2K + 1_K - r1_K \\ = (r - 1)(K - 1_K) \\ = q[(q^s - 1)/(q - 1)][K - 1_K].$$

The supposed difference set D given by (2.2) becomes

$$(3.5) \quad D = \sum_{i=1}^r H_i e_i k_i$$

in the group ring notation for ZG . Let $D(-1)$ denote the image of D under the mapping of ZG induced by the group anti-isomorphism $g \mapsto g^{-1}$, g in G .

To prove that D is a difference set in G it is sufficient to show that

$$D(-1)D = n1_G + \lambda G$$

where n and λ are those parameters of the difference set. Using (3.1), (3.2), (3.3), (3.4), (3.5) we obtain,

$$\begin{aligned} D(-1)D &= \left(\sum_i H_i e_i^{-1} k_i^{-1} \right) \left(\sum_j H_j e_j k_j \right) \\ &= \sum_i H_i^2 1_K + \sum_{i \neq j} H_i H_j e_i^{-1} e_j k_i^{-1} k_j \\ &= q^s \sum_i H_i 1_K + q^{s-1} \sum_{i \neq j} (E e_i^{-1} e_j) k_i^{-1} k_j \\ &= q^{2s} 1_E 1_K + q^s [(q^s - 1)/(q - 1)] E 1_K \\ &\quad + q^s [(q^s - 1)/(q - 1)] E(K - 1_K) \\ &= q^{2s} 1_G + q^s [(q^s - 1)/(q - 1)] G. \end{aligned}$$

Thus D is a difference set in the group $G = E \times K$ with the parameters (1.1).

4. INEQUIVALENT DIFFERENCE SETS

If D is a difference set in the group G , then Dg —defined in the group ring ZG for any group element g in G —is a difference set. We call Dg a *translate* of D . If α is an automorphism of the group G , then the image under α of a difference set in G is a difference set. Two difference sets with the same (v, k, λ, n) parameters in a group G are *equivalent* in case one can be mapped into the other by means of translations and automorphisms; otherwise, they are *inequivalent*.

The group elements $e_1, \dots, e_r$ appearing in (2.2) and (3.5) allow us to obtain some inequivalent difference sets. Define (cf. (3.5))

$$D_t = \sum_{i=1}^t H_i e_i k_i + \sum_{i=t+1}^r H_i k_i$$

for $t = 0, 1, \dots, \frac{1}{2}(q^s - 1)$, where $e_1, \dots, e_r$ are now elements of E with e_i not in H_i . We show that, if q^s is odd, then the $\frac{1}{2}(q^s + 1)$ difference sets

$$(4.1) \quad D_0, D_1, \dots, D_{(q^s-1)/2}$$

are pairwise inequivalent. Let

$$D_t' = \sum_{i=1}^t H_i e_i + \sum_{i=t+1}^r H_i$$

be the image of D_t under the natural group ring epimorphism $ZG \rightarrow ZE$ induced by the group epimorphism $G = E \times K \rightarrow E$. The coefficient of D_t' on the identity of E is $r - t = [(q^{s+1} - 1)/(q - 1)] - t$. By (3.1), the coefficients of D_t' on the non-identity elements of E cannot exceed $[(q^s - 1)/(q - 1)] + t$. Therefore, if $t \leq \frac{1}{2}(q^s - 1)$, then D_t' has a *unique* largest coefficient, namely, $r - t$. If an automorphism of K is applied to D_t , or if D_t is translated by an element of K , then the image of D_t in ZE remains unchanged. On the other hand, if an automorphism of E is applied to D_t , or if D_t is translated by an element of E , then the coefficients of the image of D_t in ZE are permuted. Assume q is a power of an odd prime. Then the order $r + 1 = q^s + \dots + q + 2$ of K is prime to the order q^{s+1} of E . Hence every automorphism of $G = E \times K$ is the product of an automorphism of E and an automorphism of K . Therefore, the property

that the image of D_t in ZE has a unique largest coefficient equal to $r - t$ is invariant under equivalence. Thus the $\frac{1}{2}(q^s + 1)$ difference sets in (4.1) are pairwise inequivalent for q -odd, as asserted.

5. HADAMARD DIFFERENCE SETS

We shall call a difference set whose parameters (v, k, λ, n) are related by $v = 4n$ a *Hadamard difference set*; see Turyn [15, pp. 329–330], who calls such difference sets *H sets*. The parameters of a Hadamard difference set are of the form (1.2)—a result first noted by Menon [9]. If $v = 4n$ for the parameters (1.1), then

$$q^s + \cdots + q + 2 = 4q^{s-1}.$$

Hence q divides 2, so $q = 2$. Conversely, if $q = 2$ in the parameters (1.1), then $v = 4n$ for every s . Thus the construction of this paper yields Hadamard difference sets with the parameters

$$(5.1) \quad (v, k, \lambda, n) = (2^{2s+2}, 2^{2s+1} - 2^s, 2^{2s} - 2^s, 2^{2s})$$

for every positive integer s . Note that, for $s \geq 2$, these difference sets can be in non-Abelian groups. Other constructions for Hadamard difference sets are given by Menon [8, 9], Turyn [15], and McFarland [7].

6. THE MULTIPLIER MINUS ONE

Suppose D is a difference set in a group G . For any integer t , let $D(t)$ denote the image of D under the mapping $g \mapsto g^t$, g in G . If $D(t)$ is a translate of D , then t is called a *multiplier* of D . Mann and McFarland [6] have shown that for each multiplier t of D there is a translate $D^* = Dg$, g in G , such that $D^*(t) = D^*$ —we say that t *fixes* the translate D^* .

We assert that the difference set D defined by (2.1) or (3.5) has minus one as a multiplier if and only if the group K is an elementary Abelian 2-group and either q is a power of 2 or for some e in E , e, e belongs to H_i for $i = 1, \dots, r$.

For, if K is an elementary Abelian 2-group and q is a power of 2, then in (3.5) we have $e_i^{-1} = e_i$ and $k_i^{-1} = k$, so $D(-1) = D$. Or, if K is an elementary Abelian 2-group and there is an e in E such that e, e belongs to H_i , then $(De)(-1) = De$.

Conversely, assume that minus one is a multiplier of the difference set D

defined by (3.5). As noted above, there is a translate Dek , with e in E and k in K , which is fixed by the multiplier minus one. Then, since E is Abelian and the elements of E commute with the elements of K ,

$$\sum_i H_i e_i e k_i k = Dek = (Dek)(-1) = \sum_i H_i (e_i e)^{-1} (k_i k)^{-1}.$$

Multiply through by H_j and use (3.2) and (3.3) to obtain

$$q^s H_j e_j e k_j k + q^{s-1} E \sum_{i \neq j} k_i k = q^s H_j (e_j e)^{-1} (k_j k)^{-1} + q^{s-1} E \sum_{i \neq j} (k_i k)^{-1}$$

Therefore,

$$(6.1) \quad H_j e_j e k_j k = H_j (e_j e)^{-1} (k_j k)^{-1}.$$

Hence $k_j k = (k_j k)^{-1}$, and this holds for $j = 1, \dots, r$. Thus r of the $r + 1$ elements of K are their own inverses. This can happen only if K is an elementary Abelian 2-group. Furthermore, (6.1) implies

$$H_j e_j e = H_j (e_j e)^{-1},$$

so

$$(6.2) \quad (e_j e)^2 \in H_j.$$

If q is a power of 2, then E is an elementary Abelian 2-group, so $(e_j e)^2 = 1$ and (6.2) holds for $j = 1, \dots, r$. If q is odd, then $h \mapsto h^2$ defines an automorphism of each H_j , so (6.2) can hold only if

$$e_j e \in H_j$$

for $j = 1, \dots, r$. This proves our assertion concerning when D has minus one as a multiplier.

Thus we can obtain difference sets having minus one as a multiplier whenever the group K is an elementary Abelian 2-group. When can this happen, i.e., for what prime powers q and positive integers s is the order of K a power of 2,

$$(6.3) \quad \frac{q^{s+1} - 1}{q - 1} + 1 = 2^u ?$$

We conjecture that the only integral solutions to (6.3) with q a prime power and s a positive integer are $q = 2$, any s and $q = 5$, $s = 2$. We have verified this conjecture for all $q^s \leq 10^4$ by some non-existence theorems for difference sets having minus one as a multiplier (unpublished) and the use of an electronic computer.

The difference sets with $q = 2$ and any s which have minus one as a multiplier are just those Hadamard difference sets discussed in Section 5 with the group $G = E \times K$ an elementary Abelian 2-group.

The case $q = 5$ and $s = 2$ is more interesting. Then we obtain a difference set having minus one as a multiplier with the parameters

$$(v, k, \lambda, n) = (4000, 775, 150, 625)$$

in the group $G = E \times K$, where E and K are elementary Abelian groups of respective orders 125 and 32. This is the first (and so far only) example of a difference set having minus one as a multiplier which is *not* a Hadamard difference set.

We now establish some special cases of the above conjecture concerning the Diophantine equation (6.3). Equation (6.3) is equivalent to

$$(6.4) \quad q^s + \cdots + q = 2^u - 2.$$

Thus if (6.3) holds for q a power of 2, then $q = 2$. And indeed (6.3) holds for $q = 2$ and every integer s , as previously noted. If q is a power of an odd prime, then (6.4) implies s is even. We show that the only solution to (6.3) with $s = 2$ and q an odd prime power is $q = 5$. This argument is due to Henry B. Mann (written communication).

If $s = 2$ in (6.3), then

$$(6.5) \quad q^2 + q + 2 = 2^u.$$

Since

$$q^2 < q^2 + q + 2 < q^2 + 2q + 1,$$

u is odd. Set $u = 2w + 1$. Then (6.5) yields

$$\frac{1}{2}q(q + 1) = (2^w - 1)(2^w + 1).$$

Clearly $w > 0$. Then $2^w - 1$ and $2^w + 1$ are relatively prime, so q must divide one of them. Thus

$$q \leq 2^w + 1.$$

But then

$$\frac{1}{2}(q + 1) \geq 2^w - 1.$$

Hence

$$2^{w+1} - 3 \leq q \leq 2^w + 1.$$

Therefore

$$2^w \leq 4.$$

Thus $w = 1, 2$ and $u = 3, 5$. Hence $q = 5$ is the only odd prime power solution to (6.5).

7. REMARKS

John F. Dillon has pointed out (private conversation) the following historical information concerning (6.5). If q is permitted to be any integer in (6.5), then solving for q leads to the equivalent Diophantine equation

$$(7.1) \quad 2^{u+2} - 7 = x^2.$$

In 1913, Ramanujan [see 12, p. 327, problem 464] conjectured that (7.1) has integral solutions only for $u = 1, 2, 3, 5, 13$. This was first proved by Nagell in 1948; see Nagell [11] for a reference to, and a translation of, his original proof. Also, see Mordell [10, pp. 205–206].

Can there exist a difference set with parameters of the form (1.1) in a cyclic group? The answer is No if the parameter k is less than or equal to 100; see Baumert [1]. Ironically, most of the seven parameter values of the form (1.1) with $k \leq 100$ are among the more difficult parameter values for which to prove the nonexistence of a difference set in a cyclic group; see Hall [3] and Baumert [1].

A difference set with parameters (v, k, λ, n) yields a more general combinatorial design which goes by the two names (v, k, λ) -*configuration* and *symmetric balanced incomplete block design*; see, e.g., Hall [4], Mann [5], or Ryser [13]. In particular, our construction yields a $(45, 12, 3)$ -configuration; see Stanton [14].

We note some further information concerning the parameters

$$(7.2) \quad (v, k, \lambda, n) = (45, 12, 3, 9).$$

There cannot exist a difference set with parameters (7.2) in a cyclic group; see, e.g., Baumert [1, p. 827, Test C]. By Section 4, there are at least two inequivalent difference sets with these parameters in the Abelian, non-cyclic group of order 45. We have established by hand computation that up to equivalence there are exactly two difference sets with the parameters (7.2) in Abelian groups.

How many inequivalent difference sets does the construction of this paper yield with given parameters in a given group? A related, but apparently more difficult question is: When does the construction yield all inequivalent difference sets with given parameters in a given group?

Note added in proof. Some of the difference sets considered herein with parameters $v = q^2(q + 2)$, $k = q(q + 1)$, $\lambda = q$, $n = q^2$ (i.e. $s = 1$ in (1.1)) were previously constructed by K. Takeuchi [on the construction of a series of BIB designs, Rep. Statist. Appl. Res., *Japan Un. Sci. Engrs.* **10** (1963), 48].

REFERENCES

1. L. D. BAUMERT, Difference sets, *SIAM J. Appl. Math.* **17** (1969), 826–833.
2. B. GORDON, W. H. MILLS, AND L. R. WELCH, Some new difference sets, *Canad. J. Math.* **14** (1962), 614–625.
3. M. HALL, JR., A survey of difference sets, *Proc. Amer. Math. Soc.* **7** (1956), 975–986.
4. M. HALL, JR., "Combinatorial Theory," Blaisdell, Waltham, Mass., 1967.
5. H. B. MANN, "Addition Theorems," Wiley, New York, 1965.
6. H. B. MANN AND R. L. MCFARLAND, On multipliers of difference sets, *Canad. J. Math.* **17** (1965), 541–542.
7. R. L. MCFARLAND, On difference sets in 2-groups, to be submitted for publication.
8. P. K. MENON, Difference sets in abelian groups, *Proc. Amer. Math. Soc.* **11** (1960), 368–376.
9. P. K. MENON, On difference sets whose parameters satisfy a certain relation, *Proc. Amer. Math. Soc.* **13** (1962), 739–745.
10. L. J. MORDELL, "Diophantine Equations," Academic Press, New York, 1969.
11. T. NAGELL, The diophantine equation $x^2 + 7 = 2^n$, *Ark. Mat.* **4** (1960), 185–187.
12. S. RAMANUJAN, "Collected Papers of Ramanujan," Cambridge University Press, Cambridge, 1927.
13. H. J. RYSER, "Combinatorial Mathematics," Wiley, New York, 1963.
14. R. G. STANTON, #7133, *Math. Reviews* **40** (1970), 1293.
15. R. J. TURYN, Character sums and difference sets, *Pacific J. Math.* **15** (1965), 319–346.